LESSON PLAN

PROPELLER THEORY (7)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	8	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Constructional details about Dornier A/C propeller, construction of blades and pitch change mechanism and spinner
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Constructional details about Dornier A/C propeller, construction of blades and pitch change mechanism and spinner
5. Specific Objectives	7.1.2.1 Trainee will be able to explain propellers and its associated maintenance practice

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of propeller construction diagrams; Safety: Ensure no loose pins or tools; Supervision: Instructor verifies diagram interpretation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an aircraft propeller?
(b) Name two types of propellers used in modern aviation.
(c) How does a propeller generate thrust?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• General construction of Dornier propeller • Blade construction and materials • Hub assembly and attachment • Pitch change mechanism components • Hydraulic pitch control system	L & Ds	PPT, IB	15 min
(b)	• Spinner construction and purpose • Blade pitch adjustment procedures • Maintenance checks for pitch mechanism • Common defects in propeller blades • Lubrication points and intervals	L & Ds	PPT, PDF	20 min
(c)	• Propeller balance and vibration causes • Inspection criteria for blade integrity • Storage and handling precautions • Record keeping for maintenance actions • Reporting defects to maintenance unit	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of the pitch change mechanism in variable pitch propellers.
(b) Describe the key maintenance checks required for propeller blades.
(c) What are the immediate actions if a propeller imbalance is detected?

12. Home Assignment and Follow Up	Prepare a brief note on the differences between fixed pitch and variable pitch propellers, referencing the maintenance practices discussed in LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Various checks on propeller on the ground during ground run	Refer to the master syllabus reference list	LP 07

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min